Think like a Coder - THE CHASM

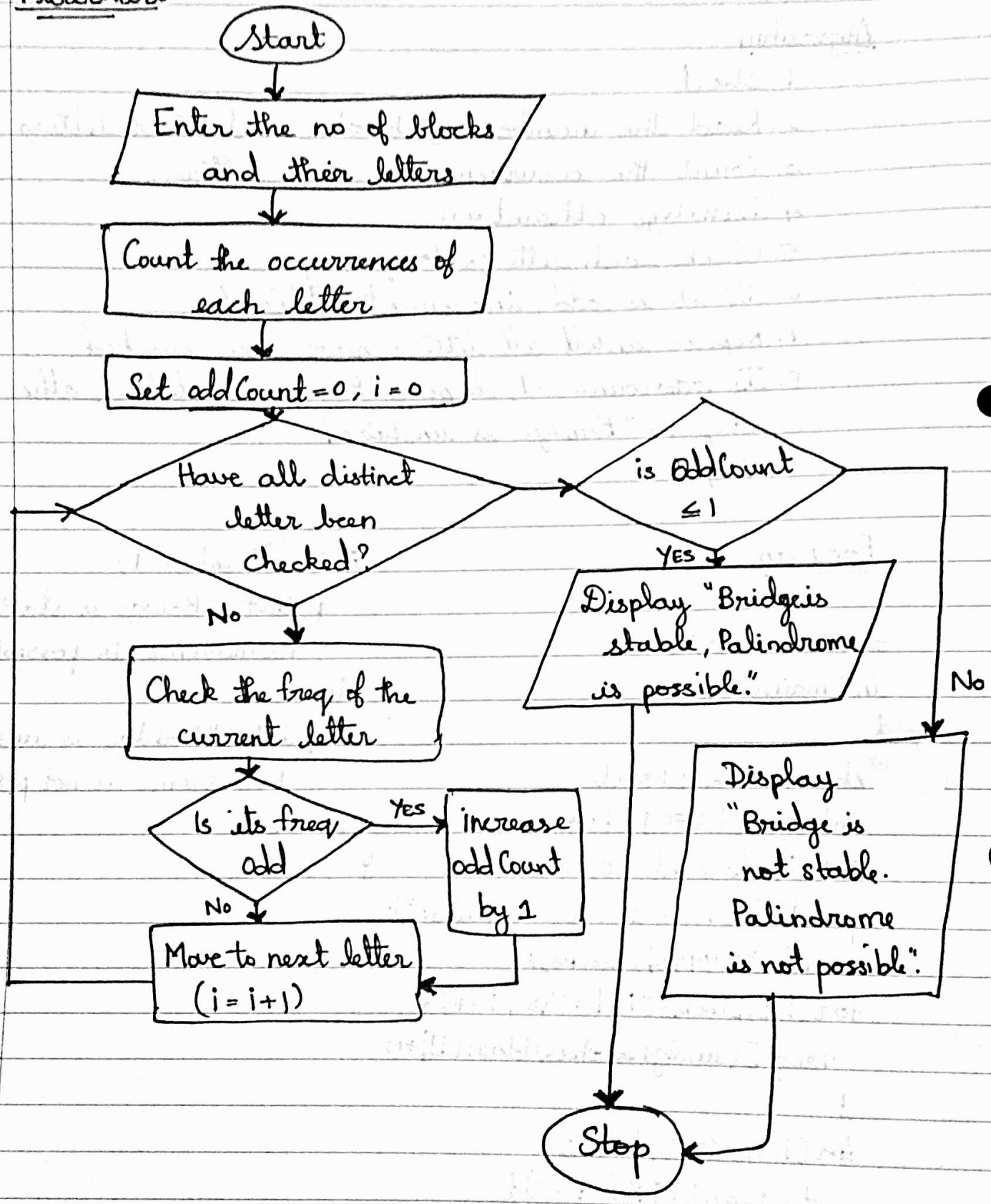
Algorithm:

1. Start
2. Read the number of blocks and their letters
3. Count the occurrences of each letter
4. Initialize oddCount = 0
5. Check each letter's frequency
6. If it is odd, increment oddCount
7. Repeat until all letters have been checked
8. If oddCount ≤ 1, display "Bridge is stable", otherwise display "Bridge is unstable."
9. Stop.

Program:

```
#include<stdio.h>
int main()
{
 char blocks[1000];
 int freq[256]={0};
 int i,oddCount=0;
 printf("Enter the block letters:");
 scanf("%999s",blocks);
 for(i=0,blocks[i]!='\0',i++){
  freq[(unsigned char)block[i]]++;
 }
 for(i=0;i<256;i++){
  if(freq[i]%2!=0){
    oddCount++;
 }}
 }
 if(oddCount<=1)
  printf("Bridge is stable.
   Palindrome is possible.");
else
  printf("Bridge is unstable.
   Palindrome is not possible.");
 return 0;
}
```

Flowchart:


Start
Enter the no of blocks and their letters
Count the occurrences of each letter
Set oddCount = 0, i = 0
Have all distinct letter been checked?
No
Check the freq of the current letter
Is its freq odd
YES
increase oddCount by 1
No
Move to next letter (i = i+1)
is oddCount ≤ 1
YES
Display "Bridge is stable, Palindrome is possible."
No
Display "Bridge is not stable. Palindrome is not possible".
Stop